

Aiyagari–McGrattan (1998) — Household Stage

Paper: S. Rao Aiyagari and Ellen R. McGrattan, “The Optimum Quantity of Debt,” *Journal of Monetary Economics* 42, 1998.

Scope: Household dynamic program only. Two variants are covered: **Model A** (inelastic labor, lump-sum taxes; §§1–9) and **Model B** (elastic labor, proportional taxes; §10 — the paper’s benchmark). Prices $(r, \tilde{w})$ and fiscal policy (τ, χ) are parameters pinned down outside this stage by general equilibrium.

1. Stage Overview

The household problem is a **single consumption-savings stage** with a persistent discrete Markov shock on labor productivity. Growth normalization (tilde variables) renders the problem stationary. The stage is EGM-amenable: CRRA utility with a single continuous control and a linear budget constraint.

Effective discount factor:

$$\tilde{\beta} := \beta(1 + g)^{1-\nu}, \quad \tilde{\beta} < 1 \text{ required.}$$

Instantaneous utility:

$$u(\tilde{c}) := \frac{\tilde{c}^{1-\nu}}{1-\nu}$$

2. Perch Structure and Information Sets

The stage has three perches. Each perch defines an information set — the set of variables the agent (or the modeler) can condition on at that point.

Arrival Perch ($\prec$)

The agent enters the period knowing assets and productivity.

Variable	Space	Description
$\tilde{a}$	$X_{\tilde{a}} := \mathbb{R}_+$	Output-normalized beginning-of-period assets
e	$\mathcal{E} = \{e_1, \dots, e_N\}$	Idiosyncratic productivity (already realized)

Information set: $\mathcal{F}_{\prec} = \sigma(\tilde{a}, e)$. Both variables are known. No decisions have been made; no new shocks resolve here.

Decision Perch ($\circ$)

The agent observes cash-on-hand (constructed from arrival states) and chooses consumption.

Role	Variable	Space	Description
state	$\tilde{m}$	$X_{\tilde{m}} := \mathbb{R}_+$	Cash-on-hand (market resources)
state	e	$\mathcal{E}$	Productivity (carried through, conditions continuation value)
control	$\tilde{c}$	$\mathbb{R}_+$	Consumption

Information set: $\mathcal{F}_\circ = \sigma(\tilde{m}, e)$. Cash-on-hand is a deterministic function of arrival states, so no new uncertainty resolves, but the *representation* of the state changes.

Why carry e separately? Cash-on-hand $\tilde{m}$ is a sufficient statistic for the *budget constraint*, but not for the *continuation value*: the conditional expectation $\mathbb{E}_{e'|e}[\cdot]$ depends on which row of Π we are in. So $(\tilde{m}, e)$ is the minimal decision-perch state.

Continuation Perch ($\succ$)

After the consumption decision, the agent holds end-of-period assets.

Variable	Space	Description
$\tilde{a}'$	$X_{\tilde{a}} = \mathbb{R}_+$	End-of-period assets
e	$\mathcal{E}$	Productivity (carried forward; conditions the Markov draw)

Information set: $\mathcal{F}_\succ = \sigma(\tilde{a}', e)$. The consumption decision has been made; $\tilde{a}'$ is determined. The next-period shock e' has **not** yet been drawn.

3. Transitions Between Perches

Arrival $\rightarrow$ Decision ($g_{\prec\circ}$)

Construct cash-on-hand from assets and the realized productivity shock:

$$\tilde{m} = (1 + r)\tilde{a} + \tilde{w}e - \tau$$

This is a deterministic, non-invertible map: many $(\tilde{a}, e)$ pairs can produce the same $\tilde{m}$, but the pair $(\tilde{m}, e)$ is recoverable given e .

Decision $\rightarrow$ Continuation ($g_{\circ\succ}$)

The budget constraint determines end-of-period assets. The paper's equation (1) (Sec. II) has $(1 + g)$ on the left-hand side:

$$\tilde{c} + (1 + g) \tilde{a}' = \tilde{m} \quad \Longleftrightarrow \quad \tilde{a}' = \frac{\tilde{m} - \tilde{c}}{1 + g}$$

with $\tilde{c} \in [0, \tilde{m}]$ (nonnegativity of both consumption and assets). Keeping the factor explicit rather than absorbing it into an effective discount factor matches the paper 1-to-1 and avoids a silent rescaling of the poststate.

Productivity passes through: e at continuation equals e at decision.

4. Movers

Backward Mover $\mathbb{B}$: Continuation $\rightarrow$ Decision (Optimization)

The backward mover solves the consumption problem, taking the continuation value $v_{\succ}(\tilde{a}', e)$ as given:

$$v(\tilde{m}, e) = \max_{\tilde{c} \in [0, \tilde{m}]} \left\{ \frac{\tilde{c}^{1-\nu}}{1-\nu} + \tilde{\beta} v_{\succ} \left(\frac{\tilde{m} - \tilde{c}}{1+g}, e \right) \right\}$$

The first-order condition picks up a $(1 + g)^{-1}$ factor from $\partial \tilde{a}' / \partial \tilde{c} = -1/(1 + g)$:

$$\tilde{c}^{-\nu} = \frac{\tilde{\beta}}{1 + g} v'_{\succ}(\tilde{a}', e).$$

Envelope condition (marginal value at decision perch):

$$v'(\tilde{m}, e) = \tilde{c}(\tilde{m}, e)^{-\nu}$$

Forward Mover $\mathbb{I}$: Decision $\rightarrow$ Arrival (Expectation)

The forward mover integrates over next-period productivity using the Markov kernel Π :

$$v_{\prec}(\tilde{a}', e) = \sum_{e' \in \mathcal{E}} \pi(e' | e) v((1 + r) \tilde{a}' + \tilde{w} e' - \tau, e')$$

Marginal (shadow) value at arrival:

$$v'_{\prec}(\tilde{a}', e) = (1 + r) \sum_{e' \in \mathcal{E}} \pi(e' | e) v'((1 + r) \tilde{a}' + \tilde{w} e' - \tau, e')$$

Stage operator: $\mathbb{T} = \mathbb{I} \circ \mathbb{B}$.

5. EGM Inversion Recipe

For each productivity state e_i and each point $\tilde{a}'_j$ on the exogenous asset grid:

Step 1 — Marginal continuation value. Compute $v'_\succ(\tilde{a}'_j, e_i)$ from the forward mover (ShadowBellman evaluated on the poststate grid).

Step 2 — Inverse Euler (InvEuler). Invert the FOC (which carries a $(1+g)^{-1}$ factor from the explicit growth-scaling of the poststate):

$$\tilde{c}_{[\succ]}(\tilde{a}'_j, e_i) = \left(\frac{\tilde{\beta}}{1+g} v'_\succ(\tilde{a}'_j, e_i) \right)^{-1/\nu}$$

Step 3 — Reverse transition (endogenous grid point). Recover cash-on-hand from the budget $\tilde{c} + (1+g)\tilde{a}' = \tilde{m}$:

$$\tilde{m}_{[\succ]}(\tilde{a}'_j, e_i) = (1+g)\tilde{a}'_j + \tilde{c}_{[\succ]}(\tilde{a}'_j, e_i)$$

Step 4 — Envelope condition. Marginal value at the endogenous decision-perch point:

$$v'(\tilde{m}_{[\succ]}, e_i) = \tilde{c}_{[\succ]}^{-\nu}$$

Output: For each e_i , the EGM produces an irregular grid of $(\tilde{m}, \tilde{c})$ pairs. Interpolation onto a regular $\tilde{m}$ -grid yields the consumption policy and marginal value.

6. Parameters

Symbol	Description	Role
β	Raw discount factor	Structural
g	Rate of technical progress	Structural
ν	CRRA coefficient	Structural
$\tilde{\beta} = \beta(1+g)^{1-\nu}$	Effective discount factor	Derived
r	Net interest rate	From GE
$\tilde{w}$	Output-normalized wage	From GE
τ	Lump-sum tax	From govt budget
Π	Markov transition matrix for e	Exogenous process

7. Shock Process

Idiosyncratic productivity e_t follows a finite-state Markov chain:

$$e \in \mathcal{E} = \{e_1, \dots, e_N\}, \quad \pi_{ij} = \Pr(e_{t+1} = e_j \mid e_t = e_i)$$

The shock is **pre-decision**: e_t is known when the agent chooses $\tilde{c}_t$. The *transition* to e_{t+1} occurs between the continuation perch of period t and the arrival perch of period $t + 1$.

8. Welfare Criterion (Steady State)

$$\Omega = \iint V(\tilde{a}, e) dH(\tilde{a}, e)$$

where $H(\tilde{a}, e)$ is the steady-state joint distribution of assets and productivity.

9. General Equilibrium Closure (Outside This Stage)

- **Firm FOCs:** $r = F_K(K, L) - \delta$, $\tilde{w} = F_L(K, L)$
 - **Government budget:** τ consistent with debt B and spending G
 - **Market clearing:** $K + B = \int \tilde{a} dH(\tilde{a}, e)$, $L = \int e dH(\tilde{a}, e)$
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10. Model B — Elastic Labor Supply, Proportional Taxes (Benchmark)

Scope notice. Model B is documented in this section for completeness. It is **not encoded** in `dolo-plus-draft.yaml` and is **not covered** by `verification.md`. This ballpark item formalizes Model A only at the executable layer; see `index.md` and `AGENTS.md` for the scope decision. Encoding Model B as a second dolo-plus stage is tracked as an out-of-scope follow-up in `AGENTS.md` “Common next tasks.”

Model B is the paper’s benchmark specification. It keeps the same three-perch structure as Model A but adds leisure $l \in [0, 1]$ as a second control and switches from lump-sum to proportional taxes (with a lump-sum transfer χ).

10.1 Objective (growth-normalized)

$$\max_{\{\tilde{c}_t, l_t, \tilde{a}_{t+1}\}} E \left[\sum_{t=0}^{\infty} \tilde{\beta}^t \frac{(\tilde{c}_t^\eta l_t^{1-\eta})^{1-\mu}}{1-\mu} \mid \tilde{a}_0, e_0 \right]$$

where $l_t \in [0, 1]$ is leisure (so $1 - l_t$ is labor supply), $\eta \in (0, 1)$ is the consumption weight in the Cobb–Douglas composite, and $\mu > 0$ is the CRRA coefficient over the composite.

10.2 Budget constraint

$$\tilde{c}_t + (1 + g) \tilde{a}_{t+1} \leq (1 + \bar{r}) \tilde{a}_t + \bar{w} e_t (1 - l_t) + \chi$$

where $(\bar{r}, \bar{w})$ are steady-state after-tax prices (so the proportional tax wedges are absorbed into the prices faced by the household) and χ is a lump-sum transfer.

10.3 Constraints

$$\tilde{c}_t \geq 0, \quad \tilde{a}_t \geq 0, \quad 0 \leq l_t \leq 1$$

10.4 Added symbols (beyond Model A)

Symbol	Role	Space / Domain	Description
l	control	$[0, 1]$	Leisure (so $1 - l$ is labor supply)
η	parameter	$(0, 1)$	Consumption weight in the Cobb–Douglas composite
μ	parameter	$\mathbb{R}_+$	CRRA coefficient over the composite
$\bar{r}$	parameter (from GE)	$\mathbb{R}$	After-tax net interest rate
$\bar{w}$	parameter (from GE)	$\mathbb{R}_+$	After-tax wage (output-normalized)
χ	parameter (from govt budget)	$\mathbb{R}$	Lump-sum transfer

All symbols from Model A (§6) are carried over unchanged except that the household faces $(\bar{r}, \bar{w})$ rather than $(r, \tilde{w})$, and the tax instrument is χ rather than τ .

10.5 Perch structure for Model B

The stage retains three perches; only the decision perch’s control dimension and the transitions change.

Arrival perch ($\prec$): same as Model A — $(\tilde{a}, e)$ with $\mathcal{F}_\prec = \sigma(\tilde{a}, e)$.

Decision perch ($\circ$): state $(\tilde{m}, e)$; controls $(\tilde{c}, l)$.

Role	Variable	Space	Description
state	$\tilde{m}$	$\mathbb{R}_+$	Cash-on-hand <i>exclusive</i> of labor income (see transition below)
state	e	$\mathcal{E}$	Productivity
control	$\tilde{c}$	$\mathbb{R}_+$	Consumption
control	l	$[0, 1]$	Leisure

Continuation perch ($\succ$): same as Model A — $(\tilde{a}', e)$.

10.6 Transitions (Model B)

Arrival $\rightarrow$ Decision ($g_{\prec\circ}$): labor income depends on the choice l , so the natural decomposition absorbs the non-labor income into $\tilde{m}$ and lets labor income enter at the decision perch via the budget:

$$\tilde{m} = (1 + \bar{r}) \tilde{a} + \chi.$$

Decision $\rightarrow$ Continuation ($g_{\circ\succ}$): the budget (with the $(1 + g)$ factor restored from §3) then determines end-of-period assets as a function of both controls:

$$(1 + g) \tilde{a}' = \tilde{m} + \bar{w} e (1 - l) - \tilde{c} \quad \Longleftrightarrow \quad \tilde{a}' = \frac{\tilde{m} + \bar{w} e (1 - l) - \tilde{c}}{1 + g}.$$

10.7 Movers (Model B)

Backward mover $\mathbb{B}$:

$$v(\tilde{m}, e) = \max_{\tilde{c} \in \mathbb{R}_+, l \in [0, 1]} \left\{ \frac{(\tilde{c}^\eta l^{1-\eta})^{1-\mu}}{1 - \mu} + \tilde{\beta} v_{\succ} \left(\frac{\tilde{m} + \bar{w} e (1 - l) - \tilde{c}}{1 + g}, e \right) \right\}.$$

At an interior optimum, the intratemporal FOC pins the consumption–leisure margin:

$$\frac{\tilde{c}}{l} = \frac{\eta}{1 - \eta} \bar{w} e.$$

Substituting $l = \tilde{c}(1 - \eta)/(\eta \bar{w} e)$ collapses the problem to a single effective control in $\tilde{c}$; the intertemporal (Inverse Euler) step then mirrors Model A’s §5 with the composite marginal utility replacing $\tilde{c}^{-\nu}$.

Forward mover $\mathbb{I}$: identical in form to Model A (§4) with $r \mapsto \bar{r}$, $\tilde{w} \mapsto \bar{w}$, and $\tau \mapsto -\chi$:

$$v_{\prec}(\tilde{a}', e) = \sum_{e' \in \mathcal{E}} \pi(e' \mid e) v((1 + \bar{r}) \tilde{a}' + \chi, e').$$

Note the *arrival-to-decision* transition above places only $(1 + \bar{r})\tilde{a}' + \chi$ into $\tilde{m}$ so labor income appears where the leisure control is actually chosen; this is one of two equivalent conventions.

10.8 EGM feasibility

EGM (§5) remains applicable via **conditional inversion**: for fixed e , the intratemporal FOC reduces the choice to $\tilde{c}$ alone; the standard EGM inversion (with the $(1 + g)$ factor of §5) then recovers the endogenous $\tilde{m}$ -grid. The composite marginal utility $u_c(\tilde{c}, l) = \eta(\tilde{c}^\eta l^{1-\eta})^{1-\mu}/\tilde{c}$ is invertible in $\tilde{c}$ at each e , so InvEuler remains closed-form.

10.5 Note on the paper’s computational method (rejected from formal DP)

The paper enforces the inequality constraints $\tilde{a} \geq 0$ and $l \in [0, 1]$ numerically with cubic **penalty terms** of the form $\zeta \cdot \min(\tilde{a}, 0)^3$ and $\zeta \cdot \min(1 - l, 0)^3$ appended to the objective. This is a **solver-level device** for the finite-element method the authors use; the formal DP uses the hard constraints above. See `verification.md` for the “rejected” status of the penalty formulation in the formalization.